Firas Al Haddad

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EDUCATION

B.Eng. Software Engineering Co-op *Expected May 2027*
Concordia University - GPA: 3.47/4.3 *Montreal, QC*

TECHNICAL SKILLS

Languages: Python, SQL, Java, TypeScript, JavaScript

Data Engineering: PySpark, Pandas, Databricks, ETL Pipelines, Medallion Architecture, Data Modeling, Parquet

Software Development: REST APIs, Asynchronous Programming, Multithreading, React, React Native, Spring Boot

Databases & BI: PostgreSQL, Power BI, DAX, Power Query (M)

Tools & Testing: Docker, Git/GitHub, Azure DevOps, CI/CD, Unit Testing, Data Validation, SonarQube, QMetry

WORK EXPERIENCE

Data Analytics Intern – Data Engineering Focus *May 2026 – Present*
Pratt & Whitney Canada *Longueuil, QC*

- Built modular Python pipelines centralizing SharePoint and API data for production Power BI dashboards.
- Implemented medallion architecture with reusable transformations, Parquet outputs, and a central orchestrator.
- Integrated Fastmarkets and Argus pricing, standardizing currencies and units for consistent metals comparisons.
- Established Azure DevOps branching and pull-request workflows; onboarded onsite and offshore teammates to Git.
- Enhanced Power BI dashboards with historical price comparisons, percentage changes, and flexible periods.

Software Engineering Intern *May 2025 – Aug 2025*
Fonex Data Systems *Montreal, QC*

- Built a Python/Tkinter desktop application to inspect and reconfigure transceivers through Buddy devices.
- Implemented asynchronous BLE communication with bleak and asyncio to read and write device memory.
- Built a multithreaded connection manager coordinating device communication with the desktop interface.
- Tested alpha and beta mobile releases using QMetry, covering device connectivity and transceiver configuration.

Data Analytics Intern *Jan 2025 – Apr 2025*
Pratt & Whitney Canada *Longueuil, QC*

- Consolidated 20+ Databricks sources into curated datasets supporting parts sales reporting in Power BI.
- Refactored PySpark ingestion, cleaning, and transformations into reusable functions for existing pipelines.
- Wrote automated tests for transformation functions within the existing testing framework.
- Redesigned dashboards to highlight daily, weekly, and monthly sales trends against business targets.

PROJECTS

Campus Guide | React Native, TypeScript, Google Maps API, Firebase *GitHub Link*

- Co-developed a mobile application supporting indoor and outdoor navigation across Concordia campuses.
- Contributed frontend interfaces connecting campus maps, directions, and calendar-based class navigation.
- Contributed automated tests within a CI/CD workflow incorporating SonarQube code-quality checks.

Focus Tracker | React, TypeScript, TensorFlow.js *GitHub Link*

- Co-developed a browser-based application estimating attention using facial landmarks and providing alerts.
- Built React interfaces for session setup, live monitoring, and post-session feedback.
- Contributed session history and Excel exports for reviewing focus metrics and survey results.